

Krishi Sahayta: A Compendium of Indian Government Policies for Farmers

****Document Version:** 1.0**

****Purpose:**** For training RAG-based AI models.

****Source Synthesis:**** Based on publicly available scheme guidelines from Ministry of Agriculture & Farmers' Welfare, Ministry of Fisheries, Animal Husbandry & Dairying, and other relevant ministries.

Page 1: Introduction & Key Objectives

****Vision for Indian Agriculture****

The Government of India has launched several flagship schemes to achieve multiple objectives:

- * ****Doubling Farmers' Income (DFI):**** A central goal to improve the economic viability of farming.
- * ****Sustainable Agriculture:**** Promoting practices that protect soil, water, and biodiversity.
- * ****Risk Mitigation:**** Providing financial security against crop loss, price volatility, and other unforeseen events.
- * ****Empowerment through Technology:**** Leveraging digital tools to improve access to information, credit, and markets.

****Core Pillars of Support:****

1. ****Income Support:**** Direct financial assistance.
2. ****Insurance & Risk Management:**** Protection from natural calamities.
3. ****Credit & Financial Inclusion:**** Access to formal loans.
4. ****Infrastructure & Post-Harvest Management:**** Building warehouses, cold chains, etc.
5. ****Input Assistance & Subsidies:**** Support for seeds, fertilizers, and equipment.

Page 2: Direct Income & Financial Support Schemes

****1. Pradhan Mantri Kisan Samman Nidhi (PM-KISAN)****

- * ****Objective:**** To provide direct income support to all landholding farmer families.
- * ****Beneficiaries:**** Small and marginal farmer families who own cultivable land.
- * ****Benefits:**** Financial assistance of ****₹6,000 per year****, payable in three equal installments of ****₹2,000**** every four months. The amount is directly transferred to the bank accounts of the beneficiaries.
- * ****Exclusion:**** Institutional landowners; farmer families holding constitutional posts; retired pensioners with a monthly pension of over ₹10,000; professionals like doctors, engineers, lawyers, etc.
- * ****Official Website:**** `pmkisan.gov.in`

****2. Pradhan Mantri Kisan Maan Dhan Yojana (PM-KMY)****

- * ****Objective:**** To provide a minimum pension of ₹3,000 per month to eligible small and marginal farmers upon attaining the age of 60 years.
- * ****Beneficiaries:**** Small and Marginal Farmers (SMFs) between 18-40 years of age.
- * ****Contribution:**** Farmers contribute between ₹55 to ₹200 per month to a pension fund, depending on their age of entry. The Central Government makes an equal matching contribution.
- * ****Payout:**** A monthly pension of ₹3,000 from the age of 60.

**Page 3: Crop Insurance & Risk Management**

****Pradhan Mantri Fasal Bima Yojana (PMFBY)****

- * ****Objective:**** To provide comprehensive insurance coverage against crop loss due to non-preventable natural risks, from pre-sowing to post-harvest.
- * ****Coverage:****
 - * ****Yield Losses:**** Standing Crops (Risk of natural fire, storm, hail, flood, drought, etc.).
 - * ****Prevented Sowing:**** Claims if a majority of farmers in an area are prevented from sowing.
 - * ****Post-Harvest Losses:**** Coverage for up to 14 days for crops dried in the field after harvest.
 - * ****Localized Calamities:**** Losses from hailstorm, landslide, and inundation.
- * ****Premium:****
 - * ****Farmers:**** Pay a very low premium of ****2%**** for Kharif crops, ****1.5%**** for Rabi crops, and ****5%**** for annual commercial/horticultural crops.
 - * ****Balance Premium:**** Paid by the Government to insurers.
- * ****Technology:**** Uses Remote Sensing, drones, and smartphones for quick assessment and claim settlement.

****Restructured Weather Based Crop Insurance Scheme (RWBCIS)****

- * ****Objective:**** To mitigate the hardship of the insured farmers against the likelihood of financial loss on account of anticipated crop loss resulting from adverse weather conditions.
- * ****How it Works:**** It uses weather parameters like rainfall, temperature, humidity, etc. If the actual weather data deviates from the pre-defined threshold, claims are automatically triggered, regardless of the actual crop yield.

**Page 4: Credit & Loan Facilities**

****1. Kisan Credit Card (KCC) Scheme****

- * **Objective:** To provide adequate and timely credit support from banks under a single window for their cultivation and other needs.
- * **Features:**
 - * **Credit for Cultivation:** Covers requirements for crop production.
 - * **Post-Harvest Expenses:** A component for produce marketing loan.
 - * **Consumption Needs:** A limited component to support family consumption needs.
 - * **Flexible Withdrawal:** ATM-enabled cards for easy withdrawal.
 - * **Interest Subvention:** A 2% interest subvention (rebate) is provided for prompt repayment. An additional 3% subvention is available for farmers repaying their loan on time, making the effective interest rate **4% per annum**.
- * **Coverage:** Now extended to Animal Husbandry and Fisheries farmers.

2. Interest Subvention for Short-Term Crop Loans

- * **Objective:** To make crop loans affordable.
- * **Benefit:** Farmers can avail of short-term crop loans of up to **₹3 lakhs** at a **7% per annum** interest rate.
- * **Additional Subvention:** Prompt repayers get an additional **3% interest subvention**, bringing the effective rate down to **4% p.a.**

Page 5: Infrastructure & Post-Harvest Support

1. Agriculture Infrastructure Fund (AIF)

- * **Objective:** To create post-harvest management infrastructure and build community farming assets.
- * **Key Features:**
 - * **Financial Support:** Provides a medium to long-term debt financing facility for investment in viable projects for post-harvest management infrastructure and community farming assets through interest subvention and financial support.
 - * **Interest Subvention:** **3% per annum** interest subvention for a loan up to **₹2 crores**.
 - * **Credit Guarantee:** Coverage for loans up to **₹2 crores** under the Credit Guarantee Fund.
 - * **Eligible Beneficiaries:** Farmers, FPOs, PACS, Agri-entrepreneurs, etc.
 - * **Eligible Projects:** Warehouses, cold chains, ripening chambers, pack houses, sorting & grading units, etc.

2. Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)

- * **Objective:** **'Har Khet Ko Paani'** (Water to every field) and improving water use efficiency through **'Per Drop More Crop'**.
- * **Components:**
 - * **Accelerated Irrigation Benefit Programme (AIBP):** To fast-track the completion of ongoing major and medium irrigation projects.

- * **Har Khet Ko Paani:** Creation of new water sources and water harvesting structures.
- * **Per Drop More Crop:** Focus on Micro-Irrigation (Drip and Sprinkler systems). Subsidies are provided to farmers for adopting these water-saving technologies.

Page 6: Soil Health & Input Management

1. Soil Health Card Scheme

- * **Objective:** To issue Soil Health Cards to farmers every 2 years so they can apply recommended doses of nutrients based on the soil test results.
- * **Process:** Soil samples are collected from farms and analyzed in labs for 12 parameters.
- * **Benefits:**
 - * Helps in judicious use of chemical fertilizers.
 - * Reduces input cost and improves soil health.
 - * Promotes balanced use of nutrients, leading to higher yield and better quality produce.

2. Paramparagat Krishi Vikas Yojana (PKVY)

- * **Objective:** To promote cluster-based organic farming.
- * **Approach:** Farmers are formed into groups and encouraged to adopt organic farming over an area of at least 20 hectares. The scheme provides financial assistance of **₹50,000 per hectare** over three years, of which **₹31,000** is given as incentive to farmers for organic inputs.

3. Neem Coated Urea

- * **Policy:** The government has mandated the neem coating of all urea supplied to farmers.
- * **Benefits:**
 - * Improves nitrogen use efficiency, reducing the required quantity of urea.
 - * Acts as a nitrification inhibitor, slowing the release of nitrogen into the soil.
 - * Checks the illegal diversion of urea for industrial use.

Page 7: Horticulture, Fisheries & Animal Husbandry

1. Mission for Integrated Development of Horticulture (MIDH)

- * **Objective:** To promote holistic growth of the horticulture sector (fruits, vegetables, root & tuber crops, mushrooms, spices, flowers, aromatic plants, coconut, cashew, cocoa, bamboo, etc.).
- * **Support Provided:** Financial assistance for:
 - * Establishment of new gardens and rejuvenation of old ones.
 - * Creation of infrastructure for post-harvest management and marketing.
 - * Adoption of technology and capacity building.

****2. Pradhan Mantri Matsya Sampada Yojana (PMMSY)****

* ****Objective:**** To bring about a Blue Revolution through sustainable development of the fisheries sector.

* ****Key Interventions:****

* ****For Marine Fisheries:**** Vessel insurance, deep-sea fishing vessels, and aquaparks.

* ****For Inland Fisheries:**** Development of fish landing centers, brood banks, and hatcheries.

* ****For Aquaculture:**** Cluster-based cage and pen culture.

* ****Post-Harvest Infrastructure:**** Cold chains, ice plants, and feed plants.

****3. National Animal Disease Control Programme (NADCP)****

* ****Objective:**** To control and eradicate Foot and Mouth Disease (FMD) and Brucellosis in livestock.

* ****Strategy:**** 100% vaccination of cattle, buffalo, sheep, goat, and pig populations.

**Page 8: Market Linkage & E-Platforms**

****1. National Agriculture Market (e-NAM)****

* ****Objective:**** To create a unified national market for agricultural commodities by integrating existing APMC (Agricultural Produce Market Committee) mandis.

* ****How it Works:**** It is a pan-India electronic trading portal which networks the existing APMC mandis to create a unified national market.

* ****Benefits for Farmers:****

* ****Transparency:**** Online auction and price discovery.

* ****More Buyers:**** Access to buyers from other states, leading to better prices.

* ****Single License:**** A single trading license valid across all e-NAM mandis.

* ****Online Payment:**** Faster and transparent payment into the farmer's bank account.

****2. Formation and Promotion of Farmer Producer Organizations (FPOs)****

* ****Objective:**** To collectively leverage farmers' production and marketing power by forming FPOs.

* ****Support:**** The government provides a financial assistance of up to ****₹18 lakhs**** per FPO for a period of 3 years for handholding, capacity building, and business planning. Credit guarantee facilities are also available.

**Page 9: State-Specific & Supplementary Schemes (Examples)**

***Note:** Your RAG system should be trained on specific state government policies. Here are examples.*

****1. (Example: State of Punjab) - Crop Diversification Programme****

* ****Aim:**** To shift area from water-intensive paddy cultivation to less water-consuming crops like Maize, Cotton, and Pulses.

* ****Incentive:**** Financial assistance per hectare for adopting alternative crops.

****2. (Example: State of Maharashtra) - Mahatma Jyotirao Phule Farmer Loan Waiver Scheme****

* ****Aim:**** To provide relief to farmers by waiving off eligible agricultural loans.

* ***(Details on eligibility and amount would be sourced from the state portal.)***

****3. PM-KISAN Beneficiary Status Check (Process)****

* Farmers can check their payment status on the PM-KISAN portal by selecting their State, entering their Aadhaar Number, Account Number, or Mobile Number.

**Page 10: How to Avail Benefits & Grievance Redressal**

****General Process for Availing Schemes:****

1. ****Aadhaar Seeding:**** Link your Aadhaar number with your bank account and land records.
2. ****Land Record Update:**** Ensure your name is correctly recorded in the land revenue records (e.g., Jamabandi, Khatauni).
3. ****Bank Account:**** Maintain an active bank account for Direct Benefit Transfer (DBT).
4. ****Approach:**** Contact the local ****Agriculture Officer****, ****Patwari****, ****Block Development Office****, or the designated nodal agency for the specific scheme.
5. ****Common Service Centers (CSCs):**** Many application forms can be submitted at nearby CSCs.

****Key Grievance Redressal Portals:****

* ****Centralized Public Grievance Redress and Monitoring System (CPGRAMS):****
`pgportal.gov.in`

* ****PM-KISAN Helpline:**** 011-23381092, Email: `pmkisan-ict@gov.in`

* ****KCC / Loan Related Queries:**** Directly at the bank branch where the account is held.

****Important Websites:****

* Ministry of Agriculture & Farmers' Welfare: `agriculture.gov.in`

* PM-KISAN: `pmkisan.gov.in`

* e-NAM: `enam.gov.in`

**Page 11: In-Depth Focus on PM-KISAN: Process & Troubleshooting**

****Detailed Operational Mechanics of PM-KISAN****

* **Registration Process:**

1. **Farmer End:** Farmers can self-register through the PM-KISAN portal or visit a Common Service Centre (CSC). Required documents include Aadhaar card, bank account details, and land record details (e.g., Khatauni, Survey number).

2. **State Government End:** The primary method is bulk upload by the State/UT Government, which verifies the farmer's details against their land revenue database.

* **E-KYC Mandate:** It is mandatory for beneficiaries to complete their e-KYC on the PM-KISAN portal. This can be done via an OTP linked to their Aadhaar number or by visiting a CSC.

* **Common Reasons for Payment Failure:**

* **Aadhaar Not Seeded/Linked:** The bank account must be linked with the Aadhaar number provided.

* **Land Record Discrepancy:** Name in bank records doesn't match land records (e.g., Singh vs Sinha, inclusion of middle names).

* **Incorrect Bank Details:** IFSC code or account number is erroneous.

* **E-KYC Pending:** The installment will be withheld until e-KYC is completed.

* **Beneficiary Status "Not Eligible":** The land records have flagged the beneficiary as falling under the exclusion category.

Grievance Redressal Mechanism for PM-KISAN:

* Farmers can use the "Beneficiary Status" and "Complaint" modules on the PM-KISAN portal, quoting their Aadhaar number or registered mobile number.

Page 12: Micro-Irrigation & Per Drop More Crop (Component of PMKSY)

Detailed Benefits and Application Process for Drip & Sprinkler Systems

* **Subsidy Pattern:**

* **Small & Marginal Farmers:** Generally, **55%** of the benchmark cost is provided as subsidy.

* **Other Farmers:** **45%** of the benchmark cost.

* **Specific States:** Some states like those in the North-East may have a higher subsidy cap.

* **Technical Units:** Assistance is provided for a minimum unit area of 0.5 hectares for Drip and 0.2 hectares for Sprinklers.

* **Application Procedure:**

1. Farmer submits an application to the District Level Nodal Agency (e.g., Agriculture Technology Management Agency - ATMA) or the designated Micro-Irrigation company.

2. A technical site survey is conducted to prepare a layout plan.

3. After approval, the farmer purchases the system from an empanelled vendor.

4. The subsidy amount is directly transferred to the farmer's bank account after installation and verification.

* **Impact:** Reduces water consumption by 30-50% and fertilizer use by 25-30%, while increasing yield by 20-50% for various crops.

Page 13: National Bee & Honey Mission (NBHM)

A Sub-Mission under Mission for Integrated Development of Horticulture (MIDH)

* **Objective:** To promote scientific beekeeping for cross-pollination and honey production, thereby increasing crop productivity and providing additional income.

* **Key Components:**

* **Distribution of Bee Colonies & Hives:** Financial assistance for the establishment of new apiaries with modern boxes (e.g., Langstroth hives).

* **Setting up of Integrated Beekeeping Development Centres (IBDCs):** For training, disease diagnosis, and breeding of queen bees.

* **Honey Extraction & Processing Units:** Assistance for setting up small-scale extraction, processing, and bottling plants.

* **Market Support:** Branding, labeling, and linkage with markets.

* **Eligible Beneficiaries:** Individual farmers, Farmer Producer Organizations (FPOs), and entrepreneurs.

* **Significance:** Bees are critical pollinators for many horticultural crops like apples, mustard, almonds, and cucurbits, directly enhancing yield and quality.

Page 14: Pradhan Mantri Formalisation of Micro Food Processing Enterprises (PMFME) Scheme

Objective: To provide financial, technical, and business support to micro food processing enterprises.

* **Key Features:**

* **One District One Product (ODOP) Approach:** Focuses on a single food product per district (e.g., Amritsar - Fish, Nagpur - Orange, Muzaffarpur - Litchi).

* **Seed Capital:** SHG members receive **₹40,000** as seed capital for working capital and small tools.

* **Credit-Linked Capital Subsidy:** **35%** of the eligible project cost with a maximum ceiling of **₹10 lakhs** per unit.

* **Support for Branding & Marketing:** Assistance for packaging, marketing, and e-commerce platforms.

* **Eligibility:** Existing individual micro-enterprises, Farmer Producer Organizations (FPOs), and Self-Help Groups (SHGs).

* **Ministry:** Ministry of Food Processing Industries (MoFPI).

Page 15: Rashtriya Krishi Vikas Yojana - Remunerative Approaches for Agriculture and Allied Sector Rejuvenation (RKVY-RAFTAAR)

****Objective:**** To strengthen the farmer's efforts by providing comprehensive support to the agriculture and allied sectors.

*** **Flexibility for States:**** Unlike many centrally sponsored schemes, RKVY-RAFTAAR gives states the flexibility to identify and fund projects based on their agro-climatic conditions and priorities.

*** **Key Funding Streams:****

1. ****Production Growth Scheme:**** For increasing the production of major food crops, horticulture, and allied sectors.
2. ****Infrastructure & Assets Scheme:**** For creating agriculture-related infrastructure.
3. ****Special Sub Schemes:**** For addressing emerging issues like crop diversification, innovation, and value addition.

*** **Example Projects Funded:****

- * Establishment of custom hiring centers for farm machinery.
- * Construction of on-farm ponds and water harvesting structures.
- * Development of seed processing units.
- * Setting up of biogas plants.

Page 16: National Mission on Oilseeds & Oil Palm (NMOOP)

****Objective:**** To achieve self-sufficiency in the production of oilseeds and reduce import dependency for edible oils.

*** **Strategies:****

*** **Area Expansion:**** Increasing the coverage of oilseeds (like Soybean, Groundnut, Sunflower) and oil palm in suitable regions.

*** **Productivity Enhancement:**** Through the distribution of high-yielding varieties, seed mini-kits, and integrated nutrient management.

*** **Oil Palm:**** A major focus on bringing new area under oil palm cultivation with a assured price mechanism for Fresh Fruit Bunches (FFB).

*** **Assistance for Oil Palm:****

- * **Planting Material:**** Substantial subsidy for oil palm seedlings.
- * **Maintenance Cost:**** Financial support for the initial 4-year gestation period before the plant starts bearing fruit.
- * **Price Assurance:**** A viability price is announced to protect farmers from market fluctuations.

Page 17: Sub-Mission on Agricultural Mechanization (SMAM)

****Objective:**** To promote the adoption of farm mechanization among small and marginal farmers and to custom hiring centers.

* ****Key Focus Areas:****

1. ****Custom Hiring Centers (CHCs):**** Financial assistance to establish CHCs for high-cost, hi-tech machinery (e.g., Combine Harvesters, Laser Land Levelers) that individual farmers cannot afford. FPOs are given priority.
2. ****Farm Machinery Banks (FMBs):**** Similar to CHCs, established at the village level.
3. ****Hi-Tech Hubs:**** For demonstrating high-technology equipment.
4. ****Promotion of Indigenous Tools:**** Financial assistance for purchasing identified tools and machines.

* ****Subsidy for Individual Farmers:**** Varies by state and machine category, typically higher for small & marginal farmers and for certain priority machines like seed-cum-fertilizer drills, zero till drills, and laser land levelers.

****Page 18: Pradhan Mantri Annadata Aay Sanrakshan Abhiyan (PM-AASHA)****

****Objective:**** To ensure that farmers growing oilseeds, pulses, and copra get a Minimum Support Price (MSP) even if market prices fall below MSP.

* ****It is an umbrella scheme with three components:****

1. ****Price Support Scheme (PSS):**** The government procures the notified commodities directly from farmers at MSP whenever market prices fall below MSP. The procured produce is maintained as a buffer stock.
2. ****Price Deficiency Payment Scheme (PDPS):**** In select states and for specific crops, farmers are paid the difference between the MSP and the actual market price (Monthly Average Price in the APMC mandi) directly into their bank accounts. The physical produce is sold in the open market.
3. ****Pilot of Private Procurement & Stockist Scheme (PPPS):**** Involves private sector in procurement at MSP.

****Page 19: National Mission for Sustainable Agriculture (NMSA)****

****Objective:**** To promote sustainable agriculture practices that are adaptable to climate change.

* ****Key Components:****

- * ****Rainfed Area Development (RAD):**** Adopting a watershed approach to conserve water and soil. Promotes integrated farming systems (e.g., crops + livestock + horticulture).
- * ****Soil Health Management (SHM):**** Works in tandem with the Soil Health Card Scheme to promote organic manure and balanced use of fertilizers.

- * **Climate Change & Sustainable Agriculture: Monitoring, Modeling, and Networking (CCSAMMN):** Focuses on climate change adaptation through vulnerability assessment and research.

- * **Specific Practices Promoted:**

- * Zero Tillage
- * Residue Management (e.g., Happy Seeder for managing paddy straw)
- * Agro-forestry
- * Water harvesting and micro-irrigation

Page 20: Mission Amrit Sarovar & Its Agri-Linkages

- * **Objective:** Although a pan-departmental mission, it has significant benefits for farmers.

- * **Aim:** To develop and rejuvenate 75 water bodies in each district of the country as part of the Azadi Ka Amrit Mahotsav.

- * **Direct Benefits for Agriculture:**

- * **Source for Micro-Irrigation:** The water from these ponds can be directly used for drip and sprinkler irrigation in surrounding farms.

- * **Groundwater Recharge:** Helps in raising the local water table, benefiting nearby wells and tubewells.

- * **Drought Proofing:** Provides a critical water source during dry spells for protective irrigation.

- * **Allied Activities:** Can be used for pisciculture (fish farming), providing an additional source of income.

- * **Implementation:** Involves collaboration between Departments of Rural Development, Panchayati Raj, Forest, Irrigation, and Agriculture.

Page 21: Kisan Rails & Kisan Udan - Logistics Support

1. Kisan Rails

- * **Objective:** To enable seamless transportation of perishable agricultural produce from production belts to consumption centers via the Indian Railways.

- * **Features:**

- * Refrigerated coaches to maintain the cold chain.
 - * Fast transit times compared to road transport.
 - * Runs on a priority time-table.
 - * Farmers and traders get a **50% subsidy** on the transportation cost for specified fruits and vegetables.

- * **Example Routes:** From Andhra Pradesh (Anantapur) to Delhi, from Maharashtra (Sangola) to Bihar.

****2. Kisan Udan****

- * ****Objective:**** A scheme by the Ministry of Civil Aviation to facilitate air transportation for agricultural produce from hilly, tribal, and North-Eastern regions.
- * ****How it Works:**** Airlines operating under the Regional Connectivity Scheme (UDAN) are incentivized to reserve space for agricultural produce.
- * ****Benefit:**** Allows farmers from remote areas to access high-value markets in metropolitan cities where they can get better prices for exotic vegetables, flowers, and fruits.

**Page 22: Digital Infrastructure - AgriStack & Other Initiatives**

****The Vision for Digital Agriculture****

- * ****AgriStack:**** A proposed collection of technology-based initiatives aimed at creating a unified platform for the agricultural sector.
 - * ****Core Concept:**** It plans to create a unique ****Farmer ID**** for each cultivator, which will link to their land records, historical cropping data, credit history, and scheme benefits.
 - * ****Potential Benefits:**** Streamlined service delivery, personalized advisories, easier access to credit and insurance, and improved supply chain efficiency.
- * ****Other Digital Initiatives:****
 - * ****Crop Survey:**** Using satellite and drone imagery for crop area estimation and damage assessment.
 - * ****mKisan Portal:**** Sends SMS advisories to registered farmers in their local language based on their crop and location.
 - * ****Kisan Suvidha App:**** Provides information on weather, market prices, dealers, and expert advisories.

**Page 23: In-Depth Focus on PM-KISAN: e-KYC, Data Reconciliation & State Role**

****Advanced Operational Framework of PM-KISAN****

- * ****The Critical Role of State Governments:****
 - * ****Data Validation:**** The State Nodal Department is responsible for authenticating land records and farmer data before uploading to the National PM-KISAN portal. This is done through the ****Land Revenue Database (LRD)**** software.
 - * ****Correction Modules:**** States have access to modules for correcting/rejecting beneficiary data for reasons like duplicate entries, incorrect Aadhaar seeding, or land ownership disputes.
 - * ****Physical Verification (PV):**** A mandatory process where local authorities (e.g., Patwari, Revenue Inspector) physically verify a percentage of beneficiaries to prevent fraudulent enrollments.
- * ****e-KYC: A Multi-Modal Process:****

- * **Online (OTP-Based):** Farmers can use their registered mobile number linked to Aadhaar for a One-Time Password.
- * **Biometric Authentication:** At Common Service Centres (CSCs), farmers can complete e-KYC using fingerprint/iris scanners.
- * **Face Authentication:** A newer, Aadhaar-based face authentication system is being rolled out for those with fingerprint wear issues.
- * **Consequences of Non-Compliance:** Instalments are withheld until e-KYC is completed. After a prolonged period, the beneficiary may be de-registered.

* **Data Reconciliation Process:**

- * **Aadhaar-PFMS Reconciliation:** The Public Financial Management System (PFMS) validates the bank account details against the Aadhaar number to ensure DBT success.
- * **NPCI Validation:** The National Payments Corporation of India (NPCI) validates the bank account and IFSC code.

Page 24: Micro-Irrigation - Component-Wise Subsidies & Technical Specifications

Detailed Breakdown of the 'Per Drop More Crop' Scheme

- * **Component-Wise Subsidy Cap (Example for a 1 Hectare unit):**
 - * **Drip Irrigation for Orchard Crops:** Benchmark cost ~₹1,00,000/ha. Subsidy for small farmers is 55% (₹55,000).
 - * **Drip Irrigation for Vegetable Crops:** Benchmark cost ~₹1,20,000/ha due to closer spacing. Subsidy for small farmers is 55% (₹66,000).
 - * **Sprinkler Irrigation System:** Benchmark cost ~₹40,000/ha. Subsidy for small farmers is 55% (₹22,000).
 - * **Rain-Guns:** For large fields, subsidy is available on rain-gun systems.
- * **Technical Specifications Promoted:**
 - * **Drip System:** Includes laterals, drippers, filters (sand/media, disc), fertigation tanks, control valves, and main/sub-mains.
 - * **Sprinkler System:** Includes pipelines, couplers, sprinkler heads, and a pumping unit.
 - * **Solar-Powered Pumps:** Can be integrated with micro-irrigation systems under the KUSUM scheme for a fully sustainable solution.
- * **Post-Installation Support:**
 - * **Training:** On system operation, maintenance, and fertigation (application of fertilizers through the irrigation system).
 - * **Warranty:** Empanelled companies must provide a minimum 2-year warranty on the system.

Page 25: National Mission on Natural Farming (NMNF)

A Paradigm Shift from Conventional to Chemical-Free Farming

* **Core Philosophy:** Promoting a chemical-free, agro-ecology-based farming system known as **Bhartiya Prakritik Krishi Paddhati (BPKP)**.

* **Key Principles:**

1. **Bijamrita:** Use of locally prepared microbial seed treatment using cow dung and urine.
2. **Jiwamrita:** Application of a fermented microbial culture made from cow dung, urine, jaggery, and pulse flour to enrich the soil.
3. **Acchadana (Mulching):** Use of soil mulch for moisture conservation and weed management.
4. **Whapasa:** A condition of reduced irrigation creating a harmonious environment of air and water in the soil.

* **Implementation Strategy:**

* Establishment of **15,000+** model natural farms, each at least one hectare in size, to demonstrate the viability of the practice.

* Formation of **Farmer Producer Organizations (FPOs)** dedicated to natural farming.

* Creation of a network of master trainers and "Prakritik Kheti Preraks" (Natural Farming Motivators).

* Focus on rainfed and low-moisture-retentive soils where the impact is most significant.

* **Financial Support:** Farmers receive financial assistance for inputs for on-farm infrastructure like vermicompost pits, and for purchasing indigenous cow breeds.

Page 26: Pradhan Mantri Kisan SAMPADA Yojana (PMKSY) - Comprehensive Overview

Scheme for Agro-Marine Processing and Development of Agro-Processing Clusters

* **Objective:** To create modern infrastructure for food processing, reduce wastage, increase processing capacity, and enhance farmer incomes.

* **Components and Subsidies:**

* **Mega Food Parks:** Creates state-of-the-art infrastructure for food processing with supply chain collection centres, primary processing centres, and a central processing centre. Subsidy is **₹50 Crores** per project.

* **Integrated Cold Chain:** Establishes a complete cold chain from the farm gate to the consumer. Includes pre-cooling, mobile cool rooms, and refrigerated vehicles. Subsidy is **₹10 Crores** per project.

- * **Creation/Expansion of Food Processing & Preservation Capacities (CEFPPC):** Supports setting up/modernizing food processing units. Subsidy is **₹5 Crores** per project.
- * **Backward and Forward Linkages:** Supports projects for connecting production clusters with the market. Subsidy is **₹3 Crores**.

- * **Impact:** Aims to leverage an investment of ₹30,000 crores, handle 334 lakh MT of agro-produce valuing ₹1,04,125 crores, and benefit 20 million farmers.

Page 27: In-Depth Analysis of the Kisan Credit Card (KCC) - Fisheries & Animal Husbandry

Extension of KCC to Allied Sectors

- * **KCC for Animal Husbandry & Dairy Farmers:**

- * **Purpose:** To meet short-term credit requirements for the maintenance of milch and other animals, including feed, fodder, insurance, and other expenses.

- * **Credit Limit:** Based on the number and type of animals (e.g., milch animals, sheep, goats). A standard cost of ₹40,000-₹50,000 per animal is often used for calculation.

- * **Repayment:** Linked to the productive life of the animal or the milk production cycle, typically 12 months.

- * **KCC for Fisheries (Kisan Credit Card for Fish Farmers - KCC-FF):**

- * **Purpose:** To meet working capital needs for fish/prawn feed, harvesters, pond maintenance, and fuel.

- * **Credit Limit:** Assessed based on the water area under culture. A standard cost of ₹1-₹1.5 lakh per hectare is often used.

- * **Insurance Cover:** The credit includes an insurance component for the stocked fish/prawn against disease and natural calamities.

- * **Simplified Application Process for Allied Sector KCC:**

- * Application through a simplified form at the bank branch.

- * Required documents: Aadhaar, land documents for the pond/shed, and in some cases, a technical feasibility report from the fisheries/animal husbandry department.

Page 28: Digital Agriculture - Drones & Remote Sensing for Farmers

Promotion of Drone Technology in Agriculture

- * **Sub-Scheme: "Drone Shakti" and Kisan Drones**

- * **Financial Assistance for Purchase:** FPOs receive a **100% grant** (up to a ceiling of ₹10 lakhs) for purchasing drones for demonstration and custom hiring.
- * **Custom Hiring Centers (CHCs):** CHCs can include drones in their fleet, making the technology accessible to small farmers.
- * **Drone Training for Youth:** The government sponsors training programs for rural youth to become drone pilots for agricultural purposes.
- * **Applications of Drones in Farming:**
 1. **Spraying (Liquid Pesticides/Fertilizers):** Drones can cover an acre in just a few minutes, using 20-30% less water and chemicals.
 2. **Mapping & Surveying:** Creating precise maps for soil health, crop health (NDVI), and yield estimation.
 3. **Seed Sowing:** Specially designed drones can shoot seed pods into the soil, useful for afforestation and difficult terrain.
- * **Regulatory Framework:**
 - * The Ministry of Civil Aviation's **Drone Rules, 2021** have simplified the process for obtaining a remote pilot license and registering drones for agricultural use.
 - * The Ministry of Agriculture has released standard operating procedures (SOPs) for the use of drones in pesticide application.

Page 29: National Mission on Edible Oils - Oil Palm (NMEO-OP)

A Special Focus on Achieving Self-Sufficiency in Edible Oils

- * **Ambitious Targets:** To bring an additional **6.5 lakh hectares** under oil palm cultivation by 2025-26, targeting a crude palm oil (CPO) production of **11.2 lakh MT** by 2030.
- * **Farmer-Centric Price Mechanism:**
 - * **Viability Price (VP):** A fixed price for Fresh Fruit Bunches (FFBs) is announced by the government for the entire life cycle of the oil palm plantation (20-25 years).
 - * **Formula Price (FP):** A reference price linked to international CPO prices.
 - * **Assured Price to Farmers:** Farmers will receive the **higher** of the Viability Price or the Formula Price, ensuring they are protected from international price volatility.
- * **Comprehensive Support Package:**
 - * **Subsidy for Planting Material:** Nearly 100% subsidy for oil palm seedlings.
 - * **Assistance for Inter-Cropping:** Financial support for inter-cropping during the initial 4-year gestation period to provide farmers with interim income.
 - * **Establishment of Seed Gardens:** To ensure the supply of quality planting material within India.
 - * **Focus Areas:** The scheme is being aggressively promoted in the North-Eastern states and the Andaman & Nicobar Islands due to their suitable agro-climatic conditions.

Page 30: Pradhan Mantri Fasal Bima Yojana (PMFBY) - Claims Process & Technology Integration

From Loss Assessment to Claim Settlement

* **Step-by-Step Claims Process:**

1. **Notification of Loss:** Farmer must intimate the bank/insurance company within 72 hours of the occurrence of a localized calamity (e.g., hailstorm).
2. **Crop Cutting Experiments (CCEs):** For widespread calamities, the government conducts CCEs to determine the average yield of the notified area (village/block).
3. **Yield Data Comparison:** The actual yield from CCEs is compared with the threshold yield (average of the last 7 years). If the actual yield is lower, a claim is triggered.
4. **Calculation:** Claim amount = (Threshold Yield - Actual Yield) × Sum Insured per hectare.
5. **Direct Transfer:** The claim amount is directly transferred to the farmer's bank account.

* **Technology Integration for Efficiency:**

* **YES-TECH (Yield Estimation System - Technology Enabled Crop Estimation):** Uses satellite and drone imagery to optimize the number of CCEs required, making the process faster.

* **Winnow App:** Used by field agents for capturing and uploading CCE data in real-time.

* **mKisan App for PMFBY:** Farmers can use this to check their application status, premium, and claim history.

* **Grievance Redressal:** A dedicated portal and a 24x7 call center for farmers to lodge complaints regarding non-payment or delayed payment of claims.

Page 31: Custom Hiring Centers (CHCs) - A Detailed Business Model

Making High-Cost Farm Machinery Accessible to Small Farmers

* **Eligibility and Formation:**

* **Primary Beneficiaries:** Farmer Producer Organizations (FPOs), Rural Entrepreneurs, and Panchayats are encouraged to set up CHCs.

* **Cluster-Based Approach:** A CHC is established to serve a cluster of villages, typically covering 800-1000 hectares.

* **Financial Assistance under SMAM:**

- * The government provides a **40% to 80%** subsidy on the cost of machinery, depending on the category of the beneficiary and the region (higher in NE/Hilly states).

- * The remaining amount can be financed through bank loans.

- * **Typical Machinery in a CHC:**

- * **Primary Tillage:** Tractor, Rotavator, MB Plough.

- * **Sowing/Planting:** Zero Till Drill, Seed Cum Fertilizer Drill, Potato Planter.

- * **Inter-cultivation & Plant Protection:** Laser Land Leveler, Power Weeder, Boom Sprayer.

- * **Harvesting & Threshing:** Combine Harvester, Reaper, Thresher.

- * **Post-Harvest:** Paddy Thresher, Maize Sheller.

- * **Rental Fixation:** The rental charges for the machinery are fixed by the District Level Committee to ensure they are affordable for small farmers while making the CHC a viable enterprise.

Page 32: Integrated Farming System (IFS) Models for Risk Diversification

A Holistic Approach to Farming for Enhanced Resilience and Income

- * **Core Concept:** Integrating multiple agricultural enterprises like crops, livestock, fishery, poultry, and apiary in a synergistic way so that the waste of one component becomes the input for another.

- * **Model for a 1-Hectare Farm (Example):**

- * **Crops (0.70 ha):** A combination of cereals (e.g., paddy), pulses (e.g., black gram), and oilseeds (e.g., mustard) in a rotation.

- * **Dairy (0.10 ha):** 2-3 milch animals. Their dung is used for biogas and manure.

- * **Fishery (0.10 ha):** A pond for fish culture, which can be irrigated using the farm's water source.

- * **Boundary Plantation (0.10 ha):** Fruit trees (e.g., Mango, Guava) and fodder crops along the boundary.

- * **Vermicompost Unit:** Utilizes crop residues and cattle dung to produce organic manure.

- * **Economic & Ecological Benefits:**

- * **Risk Mitigation:** Failure in one component (e.g., crop loss) is compensated by income from another (e.g., dairy).

- * **Reduced Input Cost:** On-farm generation of manure and fodder.

- * **Year-Round Employment:** Continuous engagement for the farm family.

- * **Soil Health Improvement:** Through nutrient recycling.

- * **Government Support:** Schemes like **RKVY-RAFTAAR** and **NMSA** provide financial assistance for establishing IFS models, including components like pond construction, poultry sheds, and biogas plants.

Page 33: In-Depth Analysis of the Agriculture Infrastructure Fund (AIF) - Eligibility & Application

Detailed Financial Structure and Lending Process

* **Interest Subvention Mechanics:**

- * The beneficiary pays the interest at a rate decided by the lending institution (e.g., 9%).
- * The Government provides a **3% interest subvention** per annum, effectively reducing the interest burden to **6%** for the farmer.
- * The subvention is available for a loan period of up to **7 years**.
- * **Credit Guarantee:** Under the Credit Guarantee Fund Trust for Micro and Small Enterprises (CGTMSE), loans up to ₹2 crore are covered, which provides comfort to banks and reduces the need for collateral.

* **Detailed List of Eligible Projects:**

- * **Primary Processing:** Warehouses, cold storages, pack houses, ripening chambers, sorting/grading units.
- * **Value Addition:** Pulses mills, oilseeds mills, flour mills, food processing units.
- * **Logistics:** Supply chain infrastructure, including e-marketing platforms and logistics hubs.
- * **Community Farming Assets:** Solar-powered agriculture assets, organic input production units.

* **Step-by-Step Application Process:**

1. **Project Report:** The beneficiary must prepare a detailed project report (DPR).
2. **Portal Registration:** Register on the AIF portal ('agriinfra.dac.gov.in') and submit the application along with the DPR.
3. **Bank Sanction:** The application is forwarded to an eligible bank for appraisal and sanction.
4. **Approval & Disbursement:** After bank sanction, the project is approved, and the loan is disbursed in phases.
5. **Claiming Subvention:** The lending institution claims the interest subvention from the government on a quarterly basis.

Page 34: State-Specific Spotlight - The Karnataka State Policy on Organic Farming

A Deep Dive into a State-Level Initiative

* **Mission: "Jaivik Krishi" (Organic Farming):**

- * **Goal:** To convert 10% of the state's net sown area to organic farming by 2026.

- * **Strategy:** Cluster-based approach, similar to PKVY, but with enhanced state support.
- * **Unique Financial Incentives:**
 - * **Input Cost Assistance:** Provides ₹10,000 per acre over three years (₹3,200 in the first year, ₹3,300 in the second, and ₹3,500 in the third) for purchasing organic inputs.
 - * **Certification Cost Reimbursement:** The state government fully reimburses the cost of organic certification for individual farmers and FPOs.
 - * **Market Linkage:** Establishment of dedicated "Jaivik" (organic) outlets in major cities and tie-ups with e-commerce platforms.
- * **Implementation Model:**
 - * **Jaivik Krishik Society:** Formation of farmer societies at the cluster level to manage input production, quality control, and collective marketing.
 - * **Bio-Fertilizer & Bio-Pesticide Production Units:** Financial aid to set up community-level units for producing low-cost organic inputs.

Page 35: Pradhan Mantri Kisan Maan Dhan Yojana (PM-KMY) - Enrollment & Pension Payout

Operational Details of the Farmer Pension Scheme

- * **Enrollment Process:**
 1. **Visit to CSC:** The farmer must visit a Common Service Centre (CSC) with their Aadhaar card and bank account details.
 2. **Biometric Authentication:** Enrollment is done through Aadhaar-based biometric verification.
 3. **Auto-Debit Consent:** The farmer gives consent for auto-debit of the monthly contribution from their bank account.
 4. **Co-Contributor Option:** In case of the farmer's death, the spouse can continue the scheme by contributing.
- * **Contribution Chart (Examples):**
 - * Entry Age 18: Monthly Contribution = ₹55
 - * Entry Age 25: Monthly Contribution = ₹76
 - * Entry Age 30: Monthly Contribution = ₹105
 - * Entry Age 40: Monthly Contribution = ₹200
- * **Payout and Exit Rules:**
 - * **Pension Start:** Upon attaining the age of 60, provided the contributions have been made regularly.
 - * **Exit before 60:** If a farmer exits the scheme before 60, their contributed amount plus interest will be returned.

* **Death of Farmer:** If the farmer dies, the spouse is eligible to receive 50% of the pension, or the total contribution with interest is paid to the nominee.

Page 36: National Mission on Agro-Forestry (Silvi-Medicine & Bamboo)

Integrating Trees into Farming Systems

* **Sub-Mission on Agroforestry (SMAF):**

* **Objective:** To encourage farmers to plant trees on farm bunds and barren land for additional income, timber, and climate resilience.

* **Support for Planting Material:** Provides subsidies for high-quality saplings of economically important trees like Teak, Sandalwood, Melia, and fruit trees.

* **Focus on Medicinal Plants (Silvi-Medicine):**

* **Cultivation Clusters:** Promotes the cultivation of high-demand medicinal plants like Ashwagandha, Satavar, Giloy, and Kalmegh.

* **Assistance Pattern:** Financial support of up to ₹50,000 per hectare for cultivation, processing, and marketing.

* **Linkage with AYUSH Industry:** Facilitates buy-back agreements between farmer clusters and pharmaceutical companies.

* **Bamboo Mission Integration:**

* Promotes bamboo as a crop on non-forest land.

* Supports the entire value chain: from establishing nurseries to setting up primary processing units for making bamboo products.

Page 37: Advanced Crop Insurance - Unified Package Insurance Scheme (UPIS)

A Comprehensive Insurance Product for Farmers

* **Objective:** To provide a single, bundled insurance package covering multiple risks under one policy.

* **Standardized Components of UPIS:**

1. **Crop Insurance (PMFBY):** Coverage for yield losses.
2. **Loss of Life/Accident Insurance:** Coverage for the farmer (₹2 lakh for accidental death/disability).
3. **Asset Insurance:** Coverage for agricultural assets like pumpsets, tractors, and implements.
4. **Student Safety Insurance:** Coverage for the farmer's children.

5. **Drought Relief:** A component for providing relief in case of adverse seasonal conditions.

- * **Pilot Districts:** Initially implemented in 45 districts across India, with plans for expansion.
- * **Advantage for Farmers:**
 - * **Simplicity:** A single point of contact for all insurance needs.
 - * **Cost-Effectiveness:** Bundled premium is often lower than purchasing separate policies.
 - * **Comprehensive Coverage:** Protects against crop failure, asset damage, and personal accident.

Page 38: Pradhan Mantri Formalisation of Micro Food Processing Enterprises (PMFME) - ODOPI Details

Deep Dive into the One District One Product Approach

- * **ODOPI Selection Criteria:**
 - * The product is chosen based on existing clusters, market demand, and export potential.
- * **Examples:**
 - * **Varanasi (UP):** Mango-based products.
 - * **Anantapur (AP):** Groundnut-based products.
 - * **Meerut (UP):** Sports goods (linking with local industry).
 - * **Jaisalmer (RJ):** Dates.
- * **Support for Individual Entrepreneurs:**
 - * **Credit-Linked Capital Subsidy:** 35% of the project cost, maximum ₹10 lakh.
 - * **Seed Capital for SHGs:** ₹40,000 per member for working capital.
 - * **Training & Capacity Building:** On food safety, packaging, and marketing.
- * **Support for FPOs/Cooperatives:**
 - * Grants for setting up common infrastructure like processing plants, packaging units, and cold storage.

Page 39: Kisan Rail & Kisan Udan - Operational Logistics & Subsidy Claim

How the Logistics Schemes Work in Practice

- * **Kisan Rail - Booking and Subsidy Process:**
 - * **Booking:** Farmers/traders can book parcel space through the Indian Railways' parcel office or dedicated freight corridors.

- * **Subsidy Claim:** The 50% subsidy on freight is not directly given to the farmer. It is provided as a **"Freight Relief"** to the Railways. The farmer pays only 50% of the freight charge at the time of booking.

- * **Real-time Tracking:** Wagons are equipped with GPS for real-time tracking, ensuring transparency.

- * **Kisan Udan - Operational Model:**

- * **Airlines' Role:** Airlines under the UDAN scheme reserve a certain capacity for agricultural produce on designated routes.

- * **Price Cap:** The freight rates for agricultural produce are capped to make it affordable.

- * **Coordination:** The Airports Authority of India (AAI) coordinates with the Ministry of Agriculture to identify and promote routes connecting production hubs with consumption centers.

Page 40: Digital Agriculture - Geo-Tagging and Land Record Modernization

Using GIS Technology for Scheme Implementation

- * **Geo-Tagging of Assets:**

- * **Mandate:** All assets created under government schemes (e.g., ponds under PMKSY, warehouses under AIF) must be geo-tagged.

- * **Process:** Field officials use a mobile app to capture the GPS coordinates and photos of the asset, which are uploaded to a central server.

- * **Purpose:** Prevents duplication, enables physical verification, and allows for impact assessment.

- * **Digital Land Records (National Land Record Modernization Programme - NLRMP):**

- * **Computerization of Records:** States are moving from physical 'Patwari' records to digital 'Record of Rights' (ROR).

- * **Aadhaar Seeding with Land Records:** This is a critical step to accurately identify the beneficiary for schemes like PM-KISAN.

- * **Integration with PM-KISAN:** The PM-KISAN portal is integrated with the state's land record database for real-time verification of ownership.

Page 41: Gender-Specific Initiatives in Agriculture

Empowering Women Farmers

- * **Mahila Kisan Sashaktikaran Pariyojana (MKSP):**

- * A sub-component of the National Livelihood Mission (NRLM), implemented by the Ministry of Rural Development.

- * **Objective:** To empower women in agriculture by making them resource persons and leaders ("Krishi Sakhis").

- * **Training:** Women are trained in sustainable agricultural practices, seed preservation, and organic farming.

- * **Community Resource Persons:** Successful "Krishi Sakhis" then train other women in their villages, creating a cascading effect.

- * **Support through FPOs:**

- * A special focus is given to promoting women-led FPOs.

- * **Subsidy:** Higher levels of financial assistance are provided for women-led FPOs under various schemes.

- * **Land Ownership Rights:** While a central policy, several states are actively promoting the joint titling of land in the name of both husband and wife to ensure women have access to credit and government benefits.

Page 42: Crisis Management & Contingency Plans for Farmers

Support During Natural Calamities

- * **National Disaster Response Fund (NDRF)/State Disaster Response Fund (SDRF):**

- * **Financial Assistance:** Farmers are eligible for financial assistance from the SDRF/NDRF in the event of a notified natural calamity (drought, flood, cyclone, etc.).

- * **Input Subsidy:** Provides up to **₹6,800 per hectare** for rainfed/unirrigated areas and **₹13,500 per hectare** for irrigated areas for crop loss.

- * **Other Assistance:** Includes compensation for loss of perennial crops, and livestock.

- * **Crop Contingency Plans:**

- * The Indian Council of Agricultural Research (ICAR) prepares district-level contingency plans for over 600 districts.

- * **Content:** These plans provide advisory on alternative crops, varieties, and agronomic practices to be adopted in case of a delayed monsoon or other weather aberrations.

- * **Dissemination:** Plans are disseminated through the 'mkisan' portal and KVKs (Krishi Vigyan Kendras).